XRPL Validator — Daily Engineering Report · 2026-10-17

Period: 2026-10-17 00:00 to 24:00 US Eastern (2026-10-17 04:00 to 2026-10-18 04:00 UTC); ledgers #107,709,340 to #107,731,615. Every count from the ledger and the validator's logs stops at #107,731,615, closed 03:59:52 UTC. Generated 2026-10-21 00:48 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records; Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data; data.xrpl.org agreement scores. Not available: none.

Status

Impaired (all causes resolved or fixed) — 13 validations of a ledger hash the network did not validate.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 22,276 of 22,276 ledgers; 13 validations did NOT match the network's validated ledger.
- **Native engine:** 0 engine receipts recorded.
- **Changes:** 0 deployments, 0 commits merged into main.
- **Amendment fixCleanup3_4_0** holds majority; activation due ~2026-10-23 09:34 UTC.

Service levels

Measure	Value
Ledgers closed by the network	22,276
Validations issued	22,276
Coverage (validations / ledgers closed)	100.000%
Validations matching the network's validated ledger hash	22,263 of 22,276
Unsigned during planned restarts	0
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%
Restarts	0
Basis: the hash the first peer announced as accepted (before cycle 183)	22,276

Measure	Value
External agreement, data.xrpl.org (UTC day 2026-10-17)	0.99827 (37 missed of 21,379)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,709,340–#107,731,615), by step:

Step	Ledgers with a receipt
Step 1 · native engine	0
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipts this day.

Incidents

INC-01 · 13 validations of a ledger hash the network did not validate

Window	2026-10-17 04:16 UTC to 2026-10-18 02:31 UTC
Ledgers	#107,709,590, #107,711,296, #107,716,877, #107,717,120, #107,717,835, #107,718,147, #107,722,691, #107,722,727, #107,724,038, #107,725,617, #107,729,287, #107,729,484, #107,730,234
Impact	the validator signed a hash the network did not validate for these ledgers; data.xrpl.org scores such ledgers missed
Cause	Before cycle 183 the validator signed the ledger hash from the first peer that announced a ledger as accepted, without confirmation. For these ledgers that peer had built a different ledger, so the validator signed a hash the network never validated.
Action	commit d540e6e — fix(validator): sign a ledger only once the trusted validators reach quorum on its hash; deployed in cycle 183 at 2026-10-18 22:18 UTC

Change log

Deployments

No deployments.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Independence program

No independence record for UTC day 2026-10-17; records begin 2026-10-20.

Network

Measure	Value
Ledgers closed	22,276 (#107,709,340–#107,731,615)
Mean close interval	3.88 s
Longest close interval	10 s (before #107,730,990)
Closes at least 15 s apart	0
Empty ledgers	0

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_4_0	2026-10-09 09:34 UTC	2026-10-23 09:34 UTC

Amendments enabled on mainnet: 97.

Open items

- Amendment fixCleanup3_4_0: activation due ~2026-10-23 09:34 UTC.

Corrections to earlier reports

None.